

Overview

Understand the architecture and capabilities of VCF across the stages in the private cloud lifecycle.

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Other Resources for Getting to Know VCF

In addition to an overview of the VCF capabilities, see [VCF Release Notes](#) and [Overview of VCF Deploy, Converge, and Upgrade](#).

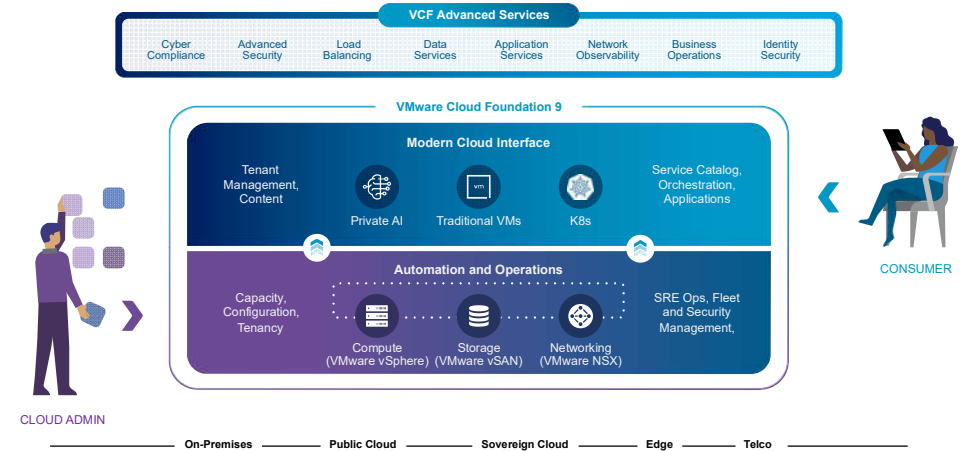
What Is VMware Cloud Foundation?

VMware Cloud Foundation (VCF) is a private cloud platform that delivers public cloud scale and agility with on-premises security, resilience and performance, while lowering total cost of ownership.

VMware Cloud Foundation[®] can help you accelerate digital innovation with faster infrastructure modernization, a unified cloud experience, along with stronger cyber resilience and platform security.

VCF delivers a consistent private cloud experience across any environment whether it's a customer managed data center, edge location or any cloud endpoint. VCF supports both traditional as well as modern workloads such as containers and AI/ML workloads, and brings modern applications faster to the market with self-service IaaS consumption experience for the application teams.

Figure 1: Modern Private Cloud Platform with VMware Cloud Foundation



- VMware Cloud Foundation is a full-stack Infrastructure as a Service (IaaS) platform that delivers software-defined compute, storage, networking, security, and management.
- Integrated automation enables a self-service platform to deploy VMs and containers for developer agility.
- Hardened platform offering built-in resilience, scaling, and clustering for non-stop operations.
- Provides cloud agility to scale infrastructure without scaling staff, delivering cloud consumption on premises.
- Provides automation and orchestration to simplify Day 0, Day 1, and Day 2 tasks.
- Simplifies full stack deployment.

What's New Information

For information about what's new in VCF 9.0, see [Release Notes](#).

VCF Components

VCF consists of the following components:

- [vSphere](#)
- [vSAN](#)
- [NSX](#)
- [VMware Cloud Foundation Operations](#)
- [VMware Cloud Foundation Automation](#)

Getting Started and a List of Capabilities

To learn how to get started, see [Getting Started with VMware Cloud Foundation](#).

For a detailed list of capabilities and links to the documentation for them, see [Capabilities by Job Function](#).

vSphere Overview

Virtualize physical infrastructure into managed CPU, storage, and networking resources.

The two core components of VMware vSphere® are VMware ESX® and VMware vCenter®. ESX is the virtualization platform where you create and run virtual machines and virtual appliances, Kubernetes workloads, and AI/ML workloads. vCenter is the service through which you manage multiple hosts connected in a network pool host resources.

vSphere manages large collections of infrastructure, such as CPUs, storage, networking, GPUs, and DPUs as a seamless and dynamic operating environment, and also manages the complexity of a data center.

Lifecycle Management of vSphere

- Lifecycle management via pre-staging ESX images, remediating hosts in parallel, and by applying updates in parallel across clusters.
- vSphere can manage infrastructure images to patch, update, or upgrade clusters using a desired state model.
- vSphere can reduce maintenance windows via reduced downtime upgrades for vCenter as well as live patching for ESX for near zero downtime for security patches.

Workload Performance

- With its GPU support, vSphere improves workload performance for modern AI workloads
- vSphere on DPUs (VMware vSphere® Distributed Services Engine™) helps accelerate infrastructure network functions on the Data Processing Unit (DPU) by utilizing available CPU cycles achieve higher workload consolidation per host.
- Distributed Resource Scheduler (vSphere DRS) enables automatic load balancing of resources allocated to workloads in a vSphere cluster. Storage DRS optimizes VM data placement as the VM is created and used over time.

Business Continuity

- High Availability with vSphere automatically restarts your VMs following physical machine failure.
- Fault Tolerance provides continuous availability of any application in the event of a hardware failure with no data loss or downtime.
- vMotion enables live migration of virtual machines with no disruption to users or loss of service, eliminating the need to schedule application downtime for planned server maintenance. Storage vMotion avoids downtime for planned storage maintenance.

Built-in Security

- Identity federation with Microsoft Entra ID (formerly Azure AD), ADFS and Okta: Secure access and account management.
- Virtual Machine Encryption: Data-at-rest encryption for virtual machine data and disks.
- vSphere Trust Authority: Remote attestation for sensitive workloads.

vSphere Documentation

For information about the built-in vSphere capabilities, see the [vSphere documentation](#).

vSAN Overview

VMware® vSAN™ aggregates local or direct-attached capacity devices of all ESX hosts in a cluster and creates a single storage pool shared across all ESX hosts in the vSAN cluster.

You can use vSphere features that require shared storage, such as vSphere HA, vSphere vMotion, and vSphere DRS, without the need for external shared storage.

Performance Across Workloads

vSAN delivers high throughput and low latency storage. For best performance, you can use vSAN Express Storage Architecture™ (vSAN ESA) with high performance NVMe-based TLC flash devices.

Agile Scaling for Deployments of All Sizes

vSAN offers flexible scaling ranging from clusters as small as two nodes (plus witness) to clusters with up to 64 nodes. Whether scaling out by adding nodes to a cluster, scaling up by adding nodes to drives, or scaling compute and storage resources independently with disaggregation, vSAN enables you to customize performance and capacity to optimize storage resources and drive down costs.

Unified Operations and Lifecycle Management

vSAN provides operational consistency from edge to cloud, with complete visibility and lifecycle management of compute and storage from a single UI.

Storage Capacity and Efficiency

Improve storage efficiency with optimized compression methods. A single-tier architecture option eliminates the need for a cache tier, further enhancing storage efficiency. vSAN enables you to store data using RAID 5 or RAID 6 for efficiency with performance at the speed of RAID 1.

Integrated File Services

vSAN enables provisioning of file shares with a single workflow and provides a unified storage control plane for both block and file storage. vSAN file services integrate Active Directory and support Kerberos network authentication and the most common protocols. vSAN file services can be used in two-node deployments and stretched cluster deployments.

VM-Centric Policy-Based Management

You can perform consistent, VM-centric operations by using policy-based management. Within the vSAN ESA, per-VM policies increase flexibility of management, enabling compression policies, or customize data services on a per-VM basis.

vSAN Documentation

For more information about the built-in vSAN capabilities, see the [vSAN documentation](#).

NSX Overview

VMware NSX® enables network connectivity, operations, and scale. It uses a software-defined approach to networking that spans data centers, clouds and application frameworks.

NSX brings networking services closer to the application, whether running on virtual machines (VMs), containers or physical servers. This allows for a cloud operating model where networks can be provisioned and managed independently of the underlying hardware.

NSX replicates the entire network model in software, enabling any network topology—from simple to complex multi-tier networks—to be created and provisioned in seconds. Users can create multiple virtual networks with diverse requirements, leveraging a combination of the services offered over NSX or from a broad ecosystem of third-party integrations to build inherently more agile and secure environments. These services can then be extended to various endpoints within and across clouds.

Layer 2 - Layer 7 Network Stack and Overlay Services

NSX provides a complete set of logical networking capabilities and services, including logical switching, routing, distributed load balancing, virtual private network (VPN), quality of service (QoS), and monitoring. These services are provisioned in virtual networks. Virtual networks are deployed non-disruptively over the networking hardware and can extend across data centers, private clouds, and container platforms.

- **Switching.** Enable logical Layer 2 overlay extensions across a routed (Layer 3) fabric within and across data center boundaries.
- **Routing.** Dynamic routing between virtual networks that is performed in a distributed manner in the hypervisor kernel, and scale-out routing with active-active failover with physical routers. Static routing and dynamic routing protocols are supported, including support for IPv6.
- **Virtual routing and forwarding (VRF).** Complete data plane isolation among tenants with a separate routing table, network address translation (NAT), and edge firewall support in each VRF on the NSX Tier-0 gateway.
- **NSX gateway.** Support for bridging between VLANs configured on the physical network and NSX overlay networks, for seamless connectivity between virtual and physical workloads.
- **VPN.** Site-to-site and unmanaged VPN for cloud gateway services.

Multi-Site Networking and Disaster Recovery

By using NSX, you can move workloads between racks and data centers by decoupling them from the underlying networks. You can use network overlays and a consistent policy framework across different infrastructures without the need for any changes to the network when moving workloads. This capability supports workload mobility for disaster recovery and workload balancing, ensuring continuous availability and optimal application performance.

- **Federation.** Centralized policy configuration and enforcement across multiple locations from a single pane of glass, enabling network-wide consistent policy, operational simplicity, and simplified disaster recovery architecture.
- **Multi-cloud networking.** Enable consistent networking and security across data center sites, and across private cloud boundaries, irrespective of underlying physical topology or cloud platform.

Network Automation with Declarative APIs

NSX can manage the provisioning, deployment, operations and retirement of networking infrastructure and applications from a central control pane. You can automate infrastructure operations and offload manual tasks with workflows and agile templating using tools such as VCF Automation, Terraform and Ansible. NSX abstracts network services from the underlying hardware, enabling programmable infrastructure through APIs. You can implement policies and workflows to automate tasks like network provisioning, security enforcement, load balancing, streamlining operations, and improving efficiency.

Self-Service Networking with VPCs and Multi-Tenancy

VPCs provide isolated network environments within a shared infrastructure. You can use Virtual Private Cloud (VPC) constructs to provision and manage network resources to business units or project teams independently, with granular control over networking policies and resource allocation.

By creating projects, you can isolate security and networking objects across tenants in a single NSX deployment. You enable multi-tenant deployment and consumption of NSX services for the provider administrator and the tenants or organizations. The provider administrator can create projects, assign users and groups, and allocate quotas to restrict the configurations available to the tenants.

NSX Documentation

For more information about the built-in NSX capabilities, see the [NSX documentation](#).

VCF Operations Overview

VMware Cloud Foundation Operations[®] (formerly VMware Aria Operations) or VCF Operations helps your organization build, manage, operate and secure your private cloud infrastructure by deploying and maintaining its fleet-level components, providing visibility and enhanced performance across the workload and infrastructure stack, and helping your organization stay compliant with regulatory standards and organizational guidelines.

VCF Operations has the following main functional areas:

- **Fleet management.** Enables operational consistency and efficient resource management of the VCF infrastructure at scale.
- **Operations management.** Provides monitoring and optimization of performance, cost and capacity, and faster troubleshooting.
- **Workload operations.** Ensures that critical applications are running as expected.
- **Performance monitoring.** Ensures applications have continuous access to resources
- **FinOps and capacity.** Helps you analyze your infrastructure expenses and optimize capacity usage.
- **Workload mobility.** Provides support for migrating and interconnecting workloads within and across VCF private cloud.
- **Security management.** Provides security management to ensure that your VCF private cloud is operationally secure.

Fleet Management

By using the fleet management functionality of VCF Operations, you can manage centrally your VCF components.

License Management

VCF provides license management in two modes - for connected environments and disconnected environments. You can upgrade your licenses and you can choose when and how to monitor VCF assets across the entire VCF stack.

For more details on license management, see the following resources:

- [Licensing](#) documentation. Explains how to use license management.
- [Licensing - Direct Customer](#) and [Licensing - Portability](#) journey maps. Represent visual guides to the sequences of license management operations.

Lifecycle Management

You can download installation and upgrade bundles, perform prechecks, upgrade the workload domains, and monitoring your upgrade operations. By streamlining this process, the environment is always running the latest and most stable software versions.

Lifecycle management simplifies and automates the process of applying updates and patches to the software components within the VCF stack, ensuring that the infrastructure complies with the latest security standards and best practices. It provides tools for planning the updates so that you can schedule maintenance windows and make changes with minimal disruption to services. It helps in managing different versions of the software components, ensuring compatibility and stability across the entire infrastructure.

For more details on lifecycle management, see the following resources:

- [Lifecycle Management](#) documentation. Explains how to use lifecycle management.
- [Manage VCF Lifecycle](#) journey map. Represents a visual guide to the sequence of lifecycle management operations.

Identity and Access Management

You can use single-source configuration in VCF Operations for identity and access management. In this way, you can use the same identity configuration across VCF components and across multiple geolocations, offering Single Sign-On (SSO) and centralized management of federated access, enhancing both flexibility and user experience.

For more details on identity and access management, see the following resources:

- [Configuring Single Sign-On](#) documentation. Explains how to use identity and access management.
- For information about the detailed sequence of operations you perform, see the [VCF Single Sign-On](#) journey map.

Certificate Management

VCF Operations introduces a unified, non-disruptive TLS certificate management system, allowing centralized control of certificates across all components in the VCF private cloud. The system provides automated workflows, integration with various certificate authorities, and auto-renewal capabilities.

Features include viewing certificates and alerts, auto-renewal for VMCA, MSCA, and OpenSSL CA, replacing default certificates with enterprise CA certificates, CA configuration and CSR generation, and certificate expiry notifications.

For more details on certificate management, see the following resources:

- [Managing Certificates in VMware Cloud Foundation](#) documentation. Explains how to use certificate management.
- [Infrastructure Management - Certificates](#) journey map. Represents a visual guide to the sequence of certificate management operations.

Password Management

Unified password management in VCF centralizes account password management of all components of the VCF private cloud. It offers a consolidated view of accounts, their statuses, and remediation needs. You can easily identify expiring or soon-to-expire passwords.

For more details on password management, see the following resources:

- [Managing Passwords for VMware Cloud Foundation Components](#) documentation. Explains how to use password management.
- [Infrastructure Management - Passwords](#) journey map. Represents a visual guide to the sequence of password management operations.

Configuration Management

Unified configuration management provides features like scheduled drift detection for vCenter and cluster objects, enabling proactive detection of configuration deviations from the desired state.

You can manage global templates across vCenter instances, integrate with Git, and schedule drift detection through the VCF Automation or VCF Operations orchestrator.

Configuration management has the following capabilities:

- Schedule drift detection. Schedule or perform on-demand/repeatable drift detection jobs with the VCF Operations console, viewing active and historical jobs.
- Download a drift status report. Generate a PDF report on template drift status for selected vCenter instances within the retention period.
- Git repository integration. Connect to a Git repository for template versioning. Git becomes the primary template owner when connection is established.
- vSphere Configuration Profile-Enabled cluster status. View aggregated vSphere Configuration Profile (VCP) drift status for activated clusters.
- Edit configuration templates. Edit templates in VCF Operations, adding, removing, or modifying vCenter Schema Configuration items with validation.

For more details on configuration management, see the following resources:

- [Scheduling Drift Detection, Configuring Source Control in VCF Operations](#) and [Managing Configuration Templates in VCF Operations](#) documentation. Explains how to use configuration management.

- [Infrastructure Management - Configuration Drift Management](#) journey map. Represents a visual guide to the sequence of configuration management operations.

Tag Management

By using unified tag management, you can manage tags across VCF components from a central location. You have centralized control of tags to resolve issues of inconsistency, scalability, and dynamic tag assignment, ensuring a uniform tagging system across the entire fleet, and simplifying management at scale.

Tag management has the following capabilities:

- Consolidated view. Obtain a single-pane-of-glass view for existing vCenter tags through their import into the common VCF Operations platform.
- Scale effectively. Scale effectively by creating new tags in VCF Operations and pushing them to multiple vCenter instances at the same time.
- Tag persistence. Maintain tag persistence during vSphere vMotion for tags originating from VCF Operations.
- Conflict resolution. Review tag conflicts across vCenter instances and learn how to resolve conflicting tags.
- Export tags. Export tags in JSON for automation purposes.

For more details on tag management, see the following resources:

- [Tags and Categories Overview](#) documentation. Explains how to use tag management.
- [Infrastructure Management - Tag Management](#) journey map. Represents a visual guide to the sequence of tag management operations.

Operations Management

VCF Operations provides capabilities designed to monitor, diagnose, and manage the health and security of your VCF environments, providing real-time insights and long-term strategic guidance.

Health and Diagnostics

Discover and remediate known issues, and monitor the operational state of the private cloud infrastructure in VCF Operations.

Health and Diagnostics Feature	Description
VCF Health	Continuously monitors the operational state of the VCF components in your environment to ensure they are functioning as expected. By proactively identifying potential issues such as expired certificates, NTP drifts, DNS misconfigurations, and configuration errors, you can take preventative actions before these issues escalate into significant problems. By using proactive monitoring, you can maintain a stable, secure, and efficient environment, reducing the risk of downtime and operational disruptions. VCF Health supports the following components and capabilities: • VCF components – vCenter – ESX – NSX – vSAN

Health and Diagnostics Feature	Description
	- VCF capabilities - vSphere vMotion - Snapshots - Workload operations - Common issues - NTP Drifts - DNS lookup issues - Certificate expiry For more details on VCF Health, see the following resources: Monitor Your Environment and Prevent Issues with VCF Health documentation. Explains how to use VCF Health. Troubleshooting and Support - VCF Health journey map. Represents a visual guide to the sequence of operations in VCF Health.
Diagnostics	Automatically detects known issues and critical vulnerabilities within the VCF software stack, providing detailed descriptions, root cause insights, and actionable remediation steps to resolve problems and mitigate risks. Diagnostic Findings The Diagnostic Findings feature offers a consolidated view of known product issues, VMSA-based security exposures, and best-practice recommendations across your VCF environment. Diagnostic Findings helps VCF administrators gain visibility into potential issues and risks, prioritize resolution efforts, and achieve uninterrupted operations. For more information how to use Diagnostics Findings, see the What are Diagnostic Findings documentation. For a visual guide to the sequence of operations in Diagnostic Findings, see the Operations - Diagnostic Findings journey map. Log Assist Log Assist streamlines the process of manually gathering and uploading support bundles used by VMware Technical Support Engineers (TSEs) to help solve customer technical issues. With the Log Assist feature, you can transfer support bundles to Broadcom Support from VCF Operations for the purpose of troubleshooting an open Support Case. For more information about how to use Log Assist, see the Use Log Assist to Accelerate Your Support Cases documentation. For a visual guide to the sequence of operations in Log Assist, see the Troubleshooting and Support - Log Assist journey map.

Integrated Operations

Integrated operations provide a complete network view with real-time health monitoring, traffic analysis, and application insights. They also simplify storage management by consolidating key insights into a unified view, enabling proactive optimization, efficient resource allocation, and minimized downtime. Data and Site Resiliency Monitoring provides centralized visibility into data protection metrics across private clouds. Log Operations introduces a standardized log format enabling different parts of the infrastructure to be troubleshooted together across different components.

Integrated Operations Features	Description
Analyze Logs	Facilitates deep dive into infrastructure logs, with powerful query and filtering options to troubleshoot issues across your virtualized environment. You can analyse and visualize log data, and create custom queries for ongoing monitoring. VCF Operations 9.0 introduces Operations for logs, a new native logging solution enabling you to configure log-based alerts, create dashboards, and explore logs directly within VCF Operations. Alerts, dashboards, and saved queries can be added to management packs. For other log configuration, or legacy log functionality, continue using the VCF Operations for logs UI. Operations for logs supports VMware Aria Operations for Logs content packs. Transition to using log-based alerts, dashboards, and saved queries using VCF Operations management packs. For more details on log analysis, see the following resources: Configuring and Analyzing Logs documentation. Explains how to perform log analysis. Operations - Logs journey map. Represents a visual guide to the sequence of log analysis operations.
Network Operations	View and manage the operational state of your VCF networking, allowing for streamlined troubleshooting and ensuring network performance is optimized. Key capabilities include: - Overview and summary of vSphere networking and NSX inventory. - Visibility into network alerts and their trends. - Monitoring of NSX health.* - Monitoring of Virtual Private Clouds (VPCs).* - Discover applications based on network flows.* - Traffic summary across the VCF environment for the past 24 hours.* *Requires VCF Operations for networks to be deployed and attached. For more details on network operations, see the following resources: Network Operations documentation. Explains how to monitor network operations. Operations - Network Operations journey map. Represents a visual guide to the sequence of network operations.
Storage Operations	View and manage centrally the operational state of your VCF storage, reviewing storage capacity and performance. Storage Operations provides the following capabilities: - Centralized view of storage inventory, capacity utilization, and key performance statistics. - Visibility into the health of vSAN clusters, and drill down experience for further analysis. - Visibility into vSAN deduplication and compression for both vSAN ESA and OSA deployments, including the following objects: - Space efficiency metrics - New deduplication dashboards - Health alerts - vSAN Performance Diagnostics, providing the following capabilities: - Improves the performance of your vSAN cluster by running vSAN performance benchmark tests for a desired performance outcome. - Maximize Throughput - Maximize IOPs - Minimize Latency - Troubleshoots performance issues and provides guided remediation of the issues. The diagnostic tests can be run on both vSAN OSA and ESA HCI architectures. For more information about how to monitor your storage, see the Monitoring Storage Operations documentation.
Management Packs	A VCF Operations management pack, is a software package that allows you to monitor, manage, and troubleshoot infrastructure by collecting metrics, events, and alerts from the integrated system.

Integrated Operations Features	Description
	- Marketplace plug-in Introducing a new Marketplace feature in VCF Operations 9.0 that allows you to find, download, and install integrations. For more information, see Adding and Managing Management Packs from VMware Marketplace. - Custom management packs You can create a custom management pack in VCF Operations. The following features are available starting with VCF 9.0: - Design, test, and install custom management packs. - Support for JSON and XML REST APIs. - Extend existing objects with new metrics and relationships. For more information, see Management Pack Builder. - Log for management packs Starting with VCF 9.0, log information added for vSphere, vSAN, VCF Operations, and VCF Automation management packs.

Workload Operations

Workload operations ensure that critical applications are running properly by monitoring its performance, availability and end-user experience. For container-based applications, VCF Operations provides native integration with vSphere Supervisor which enables automatic monitoring of Supervisor instances and vSphere Kubernetes Service (VKS) clusters at a granular level, shortening troubleshooting time.

A business application consists of a set of interconnected applications or services running on one or more virtual machines which together offer a service to users. Business applications are enhanced to include application discovery based on Network Flows.

For more information, see the [Workload Monitoring and Observability](#) documentation.

Performance Monitoring

Performance monitoring ensures applications have continuous access to resources and AI-driven troubleshooting and remediation enables faster time to resolution of issues while keeping costs down. Manage, monitor and troubleshoot your applications by collecting application metrics from your virtual machines.

Management Packs

A VCF Operations management pack, is a software package that allows you to monitor, manage, and troubleshoot infrastructure by collecting metrics, events, and alerts from the integrated system.

- Marketplace Plug-in
Starting with VCF Operations 9.0, you can download and install integrations directly from within VCF Operations. For more information, see the [Installing Integrations in a Connected Environment](#) documentation.
- Custom Management Pack
You can create a custom management pack in VCF Operations. You can perform the following tasks:
 - Design, test, and install custom management packs.
 - Support for JSON and XML REST APIs.
 - Extend existing objects with new metrics and relationships.
For more information, see the [Management Pack Builder](#) documentation.
- Log information is available for the vSphere, vSAN, VCF Operations, and VCF Automation management packs.

FinOps and Capacity

Cost management helps you analyze your infrastructure expenses. In this way, you can identify areas that are most suitable to use capacity management for optimal utilization of your resources.

FinOps

You can use the FinOps or cost management capabilities to monitor infrastructure-related costs and associated services or license expenses with greater precision. It also enables showback and chargeback capabilities, helping application teams understand and manage the cost of the infrastructure they consume.

To support this, VI administrators and provider administrators can use a range of features such as Cost Drivers for expense attribution, Pricing Policies for rate card definitions, Showback for distributing costs based on actual usage, and Chargeback for billing tenants or application teams using predefined pricing models.

Additionally, you can use the Cost Overview and Analysis capabilities to visualize, deep-dive and make sense of your infrastructure expenses across the environments.

For more information, see the [Configuring Cost](#) documentation.

Capacity Management

Capacity Management consists of 3 sets of capabilities - assess, optimize and plan.

Use capacity planning to evaluate the capacity requirements of your organization based on historical resource utilization and real-time predictive projections to plan ahead.

Capacity assessment lets you evaluate the available and consumed capacity (compute and storage).

Capacity optimization helps you right-size or reclaim any underutilized or wasted resources.

- Workload Placement lets you optimize virtual machines and storage across clusters to reduce resource contention and maintain optimum system performance.
- By using What-If Analysis, you can plan for an increase or decrease in workload or capacity requirements in your virtual infrastructure. To evaluate the demand and supply for capacity on your resources, and to assess the potential risk to your current capacity, you can run scenarios for adding and removing workloads. You can also determine how much capacity you require to make a migration work. You can run one scenario or group scenarios and run them cumulatively.

For more information, see the [Viewing Object Capacity in the Capacity Tab](#), [What-If Analysis - Workload Planning: Traditional](#) and [What-If Analysis: Modeling Workload, Capacity, or Migration Planning](#) documentation.

Green Score

The VMware Green Score is used to measure sustainability in digital operations through data center virtualization. Green Score consists of multiple parameters for effective measurement of the score. Virtualization plays a crucial role in optimizing resource utilization, helping reduce hardware footprints in the data centers and further enabling savings in power consumption and floor space. This improves the overall efficiency and curbs carbon emissions resulting from IT infrastructure and operations.

For more information, see the [Configuring Green Score](#) documentation.

Workload Mobility

VCF Operations HCX is a workload mobility platform for migration and interconnection of workloads within and across VCF private clouds.

You can perform secure and automated migrations without requiring changes to applications or network configurations. Important features include multiple migration types, such as live and bulk migration, as well as network extension, which preserves existing IP addresses and network topologies. Thus, you can place workloads based on business requirements. The platform minimizes downtime during migrations, improves scalability, and supports compliance and security throughout workload transitions.

For more information, see the [Workload Mobility](#) documentation.

Security and Compliance

The security management area of VCF Operations includes capabilities such as Security Operations, Event Auditing, and Compliance Checks.

Security Operations

The Security Operations dashboard is designed to provide a comprehensive view of your security posture in VCF. This dashboard consolidates elements related to user and infrastructure security to help you identify areas that might require attention and tightening based on your security exposure. It complements the Compliance capability within VCF Operations by surfacing core aspects of VCF security in one place.

For more details on security operations, see the following resources:

- [Monitoring Security Operations](#) documentation. Explains how to use security operations.
- [Operations - Security Operations](#) journey map. Represents a visual guide to the sequence of security operations.

Events Auditing

Event auditing provides visibility into platform interaction changes across the VCF infrastructure and insights into suspicious access events and policy violations. It increases user accountability as each action is registered in these events.

For more information, see the [Viewing and Configuring Audit Events](#) documentation.

Compliance

Ensure compliance for your VCF infrastructure and cloud environments against optimum configurations, maintaining a consistent configuration across SDDC components, reducing organizational and business risk.

For more information, see the [Measuring Compliance of Objects](#) documentation.

VCF Automation Overview

VMware Cloud Foundation Automation® (formerly VMware Aria Automation) or VCF Automation enables IT teams and cloud service providers (CSPs) to deliver a self-service private cloud for AI, Kubernetes, and VM-based applications. The solution simplifies the process of provisioning and scaling a multi-tenant private cloud with out-of-the-box Infrastructure as a Service (IaaS) offerings, accelerating the time to market for applications while maintaining control using policy-based governance.

VCF Automation has the following capabilities:

- [Cloud Services](#). Provides a set of cloud services to meet the needs of application teams.
- [Provider Management](#). Supports IT teams and cloud service providers to effectively manage, deliver, and scale cloud services across multiple vCenter and VCF instances.
- [Organization Management](#). Provides capabilities that enable the organization administrator of a Line of Business or a tenant to further organize and govern resources allocated to them among application teams.

- [vSphere Supervisor](#). Provides the capability to run Kubernetes workloads directly on ESX hosts and to create upstream Kubernetes clusters within dedicated namespaces.

For a visual guide to the sequence of operations for transitioning from VMware Aria Automation to VCF Automation, see the [Holistic Aria Automation to VCF Automation](#) journey map.

Cloud Services

VCF Automation provides a set of cloud services which are accessible on a self-service basis using multiple interfaces, including UI, CLI, and API.

With this interface, users can provision the following types of workloads:

- VMs
- Kubernetes workloads
- Networking
- Volumes
- Secret stores
- Databases
- Harbor container registries
- External DNS
- Certificates
- AI workloads

Additionally, application teams can also provision resources on a self-service basis from Catalogs which are backed by pre-configured workload templates. These self-service options help developers focus on application development instead of managing the underlying infrastructure.

For more information about cloud services, see the [Building Cloud Applications](#) documentation.

Provider Management

IT teams and CSPs can model multi-tenancy by using the concept of Organizations in VCF Automation, effectively allocating resources and managing users specific to each Organization.

This capability provides isolated environments with dedicated network setup. IT Teams and CSPs can monitor resource usage, control permissions, and provide managed services while ensuring security through comprehensive network isolation.

For more details on provider management, see the following resources:

- [Provider Management](#) documentation. Explains how to use provider management.
- [VCF Consumption - Provider Management](#) journey map. Represents a visual guide to the sequence of provider management operations.

Organization Management

VCF Automation improves governance for users and resources by enabling organization administrators create and assign projects and vSphere Namespaces tailored for different application teams.

Organization administrators can set resource limits and policies while delegating network management through Virtual Private Clouds (VPCs) for application teams. Additionally, Policy as Code enables organization administrators to enforce instance-centric policies for Infrastructure as a Service (IaaS) resources from a central location for compliance and reducing the risk of human error.

For more details on organization management, see the following resources:

- [Organization Management](#) documentation. Explains how to use organization management.
- [VCF Consumption - Tenant Experience](#) journey map. Represents a visual guide to the sequence of organization management operations.

vSphere Supervisor

VMware vSphere Supervisor[®] is the foundation of VCF consumption and can be enabled during workload domain creation or directly in vCenter.

It transforms vSphere clusters into modern application platforms by embedding a platform-level Kubernetes control plane directly into ESX hosts, exposing a unified declarative API surface. This integration allows end users to provision and manage workloads, including VMs, containers via vSphere Pods, full Kubernetes clusters through VKS, OCI image registries, and data services through a consistent API interface.

vSphere Supervisor maintains enterprise-grade governance, security, and operational consistency through vCenter, VCF Operations, and VCF Automation. Its vSphere Namespaces provide the foundation for multi-tenancy, enabling GitOps-based workflows, and delivering cloud-like self-service in private data centers.

For an overview of vSphere Supervisor concepts and how they apply in VCF Automation, see the [vSphere Supervisor Platform](#) and [Building Cloud Applications](#) documentation.

Considerations

- Use NSX Virtual Private Clouds (VPC) workload networks
This configuration is a requirement for using VCF Automation.
While other workload network types, such as vSphere Distributed Switch and NSX Tier-0/Tier-1 segments, are supported if you plan to directly use vSphere Supervisor for application provisioning, the use of VPC workload networks gives you the ability to provide application provisioning in VCF Automation, keeping the data and infrastructure of organizations isolated.
- Determine if a Single-Zone Supervisor or Multi-Zone Supervisor deployment is required
In VCF 9.0, a Supervisor deployed in a single-zone configuration cannot be converted to a multi-zone deployment after enablement. You must determine the target deployment model before initiating the Supervisor enablement process.

Getting Started

- Single-Zone Deployment
A Single-Zone Supervisor deployment uses a single vSphere cluster to host both workloads and Supervisor management components. It can be enabled either during the [workload domain creation process](#) or via the [Supervisor Enablement workflow in vCenter](#). Additional vSphere Zones can be added after initial enablement. However, these zones are used exclusively for workload placement and do not host Supervisor control plane components.
- Multi-Zone Deployment
A Multi-Zone Supervisor deployment uses three vSphere clusters, placing each cluster into a vSphere Zone. vSphere Zones are used by workloads and Supervisor management components to deliver high availability and expose each cluster as an independent, consumable availability zone. This configuration provides a resilient, HA-capable platform. Additional vSphere Zones can be added after initial enablement. However, these additional zones are available for workload consumption only, not for Supervisor control plane components.
- Single Host
Also a Single-Zone Supervisor with a single Control Plane VM deployed to a single ESX host, with vSphere HA disabled. While it shares the same characteristics as other single-zone deployments, in this configuration, any host failure results in downtime for the Supervisor Control Plane and potentially for associated workloads.

- Easy Supervisor
This is a Single-Zone simplified deployment that uses workload networking on top of vSphere Distributed Switch, designed as a fast and simple starting point for proof-of-concept environments and production deployments leveraging vSphere Distributed Switch. For more information, see [Deploying Supervisor with a Simplified Deployment Flow](#).
An Easy Supervisor deployment consists of:
 - One Supervisor Control Plane VM
 - A shared network for both workload and management traffic
 - No load balancer
 You can deploy only VMs. Additional services, such as VKS, can be enabled by configuring a load balancer after the initial deployment.
You cannot use this deployment option with VCF Automation.

What Is vSphere Foundation?

VMware vSphere Foundation (vSphere Foundation) is an enterprise-grade workload platform for modern infrastructure delivering virtualized benefits, simplified management, cost efficiency, scalability and serves as the core foundation for VCF.

vSphere Foundation provides the following capabilities:

- Unified workload platform for running VMs and containers side by side in a native Kubernetes runtime.
- Intelligent operations management for visibility and infrastructure optimization.
- Hyperconverged infrastructure integrating compute and storage virtualization for resource management.
- Simplified deployment and scalability enabling faster application delivery and future ready infrastructure.

vSphere Foundation does not include the cloud management and integrated automation capabilities that are available in VMware Cloud Foundation. See [What Is VMware Cloud Foundation?](#)

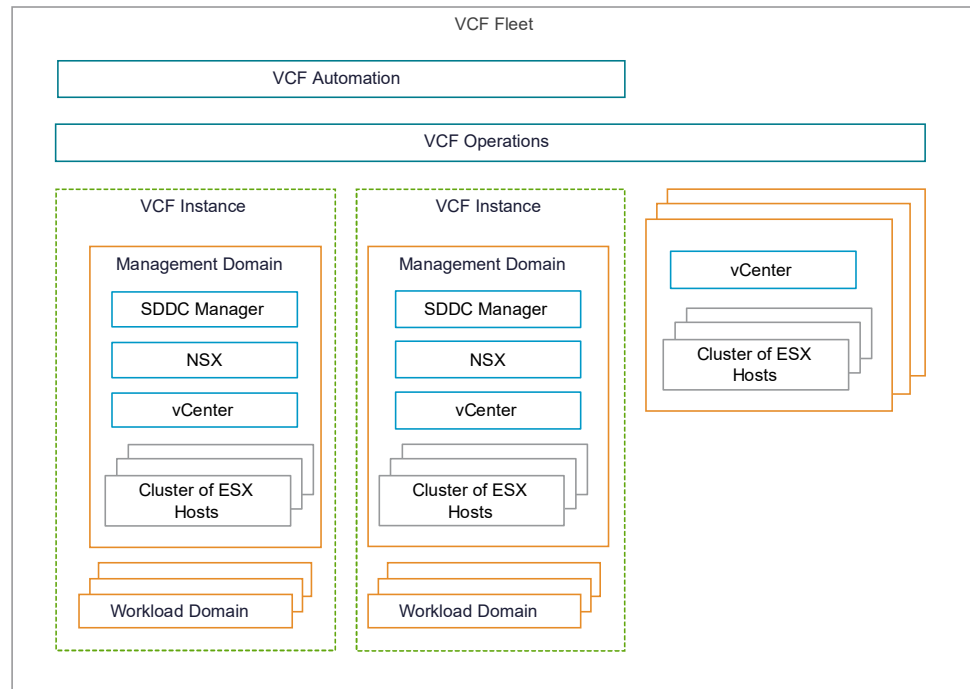
For a feature comparison and differences between VMware Cloud Foundation and VMware vSphere Foundation, see [Feature Comparison & Upgrade Paths](#) on vmware.com.

For documentation, see the [vSphere documentation](#) and [VCF Operations Overview](#).

VCF Taxonomy

The VCF fleet, VCF Instance, and VCF domains logical constructs provide a modular approach to building and managing your private cloud. Each construct has clear scope and components.

For information on designing a VCF platform according to industry recommendations and best practices, see the [VCF Design](#) documentation.

Figure 2: VCF Fleet, VCF Instances, and VCF Domains**Table 1: Constructs of a VCF Private Cloud**

Construct	Description	Components
VCF Instance	Compute, storage, networking virtual infrastructure that runs business workloads. A VCF Instance consists of VCF domains - a management domain and, optionally, additional workload domains. In a clean VCF deployment, the first vSphere cluster of a management domain runs the management components for the VCF Instance. In an environment where multiple vCenter instances have been converged to VCF, the management components can also be in vSphere clusters in the workload domains.	- SDDC Manager - NSX - vCenter - ESX
VCF fleet	An environment that is managed by a single set of fleet-level management components - VCF Operations and VCF Automation. A VCF fleet contains one or more VCF Instances and might contain one or more standalone vCenter instances, managed by the VCF Operations instance for the fleet.	- One or more VCF Instances - One or more standalone vCenter instances - VCF Operations - VCF Automation

Construct	Description	Components
	The management domain of the first VCF Instance in the VCF fleet hosts the fleet-level management components.	
VCF private cloud	The highest level of management and consumption for the underlying software-defined data center resources. A VCF private cloud can contain one or more VCF fleets.	One or more VCF fleets

VCF Domains

A VCF domain in a VCF Instance consists of the following components:

- One vCenter instance.
- One or more vSphere clusters with vSphere HA and vSphere DRS enabled.
- At least one vSphere Distributed Switches per cluster for system traffic and NSX segments for workloads.
- Dedicated NSX Manager instance, either specifically for the VCF domain or shared between VCF domains, for configuring and implementing software-defined networking.
- Optional. One or more NSX Edge clusters, added after you create the workload domain, that connects the workloads in the workload domain for logical switching, logical dynamic routing, and load balancing.
- One or more shared storage allocations as principal or supplemental storage.

Table 2: Types of VCF Domains

VCF Domain Type	Description
Management domain	The management domain is created during the deployment or convergence process by the VCF Installer. It contains the management components of the VCF Instance and, for the first VCF Instance, VCF Operations and VCF Automation. In addition to the components that are common for each VCF domain, the management domain includes the SDDC Manager appliance and the fleet management appliance, that adds workflows for infrastructure management automation to the VCF Operations instance for the VCF fleet.
Workload domain	You create workload domains in VCF Operations to run consumer workloads. For the first workload domain in your environment, VCF Operations deploys a vCenter instance and an NSX Manager instance in the management domain. For each additional workload domain, VCF Operations deploys an additional vCenter instance. New workload domains can share the same NSX Manager cluster with an existing workload domain or you can deploy a new NSX Manager cluster. Workload domains cannot use the NSX Manager cluster for the management domain.

Capabilities by Job Function

The VMware Cloud Foundation platform provides capabilities for all stages in the private cloud lifecycle and for all job functions working with the cloud's resources.

Job Functions Supported in a VMware Cloud Foundation Platform

The following functions are related to a VCF private cloud instance:

Table 3: Job Functions in VCF

Job Function	Description	
Virtual infrastructure administrator (VI administrator) or provider administrator	Build, manage and operate cloud infrastructure to provide resources for running applications.	
Platform operator	Track and analyze performance and behavior of workloads.	
Organization administrator or tenant administrator	Enable consumers to operate autonomously and efficiently in the cloud resources allocated to the organization.	
Platform engineer	Manage, automate, and monitor cloud infrastructure for self-service workload provisioning.	
End users	Application developer	Develop and deploy VM- or container-based applications in the cloud
	DevOps engineer	Manage the Kubernetes and VM infrastructure on top of the vSphere Supervisor platform in VCF for continuous integration and continuous delivery of applications.
Roles related to special infrastructure	Storage administrator	Build and manage all centralized storage in the private cloud.
	Network administrator	Build and manage network services.

VCF Capabilities

You can access VCF capabilities that are specifically designed for your job function without the need to apply expertise outside of your knowledge domain.

Table 4: Capabilities for VI Administrators

Area	Capability
Build	- Deploy a new VCF platform - Converge existing virtual Infrastructure to a VCF platform - Upgrade to VCF 9.0
	- Deploy a workload domain - Import an existing vCenter instance as a workload domain - Add a vSphere cluster or an ESX host to a workload domain
	Configure a vSAN stretched cluster
	Manage network connectivity using NSX Edge clusters and transit gateways in vCenter
	Manage networking using virtual private clouds in vCenter
	Manage virtual infrastructure in vCenter
Manage	Licensing
	Configure identity and access using VCF Single Sign-on
	Link vCenter instances for centralized inventory management
	Manage certificates across the VCF fleet
	Manage passwords across the VCF fleet
	Manage tags across the VCF fleet
	Manage the configuration settings of vCenter instances and vSphere clusters

Area	Capability
Operate	Link vCenter instances for centralized inventory management
	Back up and restore the VCF fleet and its managed VCF Instances
	Migrate workloads across workload domains and vCenter instances
	Configure alerts
	Perform log analysis and troubleshooting
	Monitor network operations
	Monitor storage operations
	Perform infrastructure diagnostics and VCF health monitoring
	- Configure and use monitoring dashboards - Configure and use monitoring reports
	Perform workload monitoring and observability
Consume	Manage cost and capacity
	Mitigate security risks and discover objects that are not compliant with a security standard
	Deploy and configure vSphere Supervisor for running Kubernetes and VM workloads
	Manage infrastructure, and identity and access for consumer organizations

Table 5: Capabilities for Storage Administrators and Network Administrators

Area	Capabilities
Build	Manage vSAN storage
	Manage networking in NSX

Table 6: Capabilities for Platform Operators

Area	Capabilities
Manage	Link vCenter instances for centralized inventory management
	Manage certificates across the VCF fleet
	Manage passwords across the VCF fleet
	Manage tags across the VCF fleet
	Manage the configuration settings of vCenter instances and vSphere clusters
	Link vCenter instances for centralized inventory management
	Back up and restore the VCF fleet and its managed VCF Instances
Operate	Migrate workloads across workload domains and vCenter instances
	Configure alerts
	Perform log analysis and troubleshooting
	Monitor network operations
	Monitor storage operations
	Perform infrastructure diagnostics and VCF health monitoring
	- Configure and use monitoring dashboards - Configure and use monitoring reports
	Perform workload monitoring and observability

Area	Capabilities
	Manage cost and capacity Mitigate security risks and discover objects that are not compliant with a security standard

Table 7: Capabilities for Organization Administrators and Platform Engineers

Area	Capabilities
Consume	Prepare the collection of VM-based workload blueprints and infrastructure services of your organization for end users Prepare the collection of VM- and container-based workload blueprints and infrastructure services of your organization for end users.

Table 8: Capabilities for DevOps Engineers and Application Developers

Area	Capabilities
Consume	Provision VMs Provision Kubernetes clusters Provision vSphere Pods Provision persistent volumes Provision secret stores Install Harbor container registries

Getting Started with VMware Cloud Foundation

You start building a VCF platform by deploying a new platform, converging your existing virtual infrastructure to VCF, or upgrading existing your VCF platform. After you deploy your VCF Instances and VCF fleet, you can continue with adding workload domains and integrating them in VCF Automation for end-user consumption.

In this article:

- [Deploy a VCF Fleet or a VCF Instance](#)
- [Add to or Move VCF Instances Between VCF Fleets](#)
- [Add a vCenter Instance to the VCF Fleet](#)
- [Complete the Configuration of the VCF Fleet, VCF Instances, and Workload Domains](#)
- [Add Workload Domains to Your Cloud Organization Structure](#)
- [Detailed Journeys](#)
- [Design Blueprints](#)
- [List of VCF Capabilities](#)
- [What to Do Next](#)

Deploy a VCF Fleet or a VCF Instance

The first step in building a VCF platform is deploying a VCF fleet or a VCF Instance in a VCF fleet.

Operation Type	Options
Deploying a New VMware Cloud Foundation or vSphere Foundation Private Cloud.	• Deploy a new VCF fleet. You deploy the components of the VCF fleet and of the first VCF Instance with the initial management domain. In addition to the management components of the VCF Instance, the initial management domain hosts the components of the VCF fleet too. • Deploy a new VCF Instance within an existing VCF fleet. You deploy a VCF Instance with its management domain.
Converging Your Existing Virtual Infrastructure to VMware Cloud FoundationPlatform.	• Converge to a new VCF fleet. You can use VMware Aria Operations and other VMware Aria Suite components in your environment as the VCF fleet. The vCenter instance and ESX hosts running VMware Aria Operations become the first VCF Instance with the initial management domain. If your environment does not have VMware Aria Operations, then you can make a vCenter and its managed hosts the first VCF Instance and deploy VCF Operations and the other required fleet components on it. • Converge to a new VCF Instance within a VCF fleet. You can add a vCenter and its managed ESX hosts as a VCF Instance with its management domain that is managed by an existing VCF fleet.
Upgrade Your Management Domain to VCF.	• Upgrade the existing VMware Aria components and management domain components. Then, deploy the rest of the VCF fleet components that you need. • If VMware Aria Operations is not included in your VCF 5.2 environment, deploy VCF Operations and then VCF Operations fleet management. Then, upgrade the the other VMware Aria components in your environment and the management domain. Deploy the rest of the VCF fleet components that you need.

Add to or Move VCF Instances Between VCF Fleets

After you deploy the VCF fleet, you can add existing VCF Instances running VCF 5.x or later to the fleet or move them between fleets. See [Add, Remove, or Move VCF Instances across VCF Fleets](#).

Add Workload Domains to a VCF Instance

After you deploy the VCF fleet and VCF Instances, you can start adding workload domains with infrastructure for consumer workloads to the VCF Instances in one of the following ways:

- [Deploy new workload domains.](#)
As part of workload domain deployment, you can enable vSphere Supervisor with NSX VPC on the initial cluster of the workload domain. You can then add the initial infrastructure of the workload domain to VCF Automation.
- [Import vCenter instances, managing consumer workloads, as workload domains.](#)
You can import components that are compatible with VCF 5.x and upgrade to version 9.0 after you import them as a workload domain.

Add a vCenter Instance to the VCF Fleet

You can add virtual infrastructure under a vSphere Foundation license to a VCF fleet. See [vSphere as a data source in VCF Operations](#).

Complete the Configuration of the VCF Fleet, VCF Instances, and Workload Domains

After you deploy the VCF fleet or VCF Instance or add a workload domain to a VCF Instance, complete its configuration by performing the following tasks. Assigning a license must be done right after you complete deployment.

- [Assign a license to the VCF fleet, VCF Instance or workload domain.](#)

- [Configure depot settings.](#)
- [Deploy VCF Identity Broker and configure identity and access management.](#)
- [Configure backup of the management components.](#)
- [Configure certificate management in VCF Operations.](#)
- [Configure password management.](#)
- [Add clusters and individual ESX hosts to your VCF domains.](#)

Add Workload Domains to Your Cloud Organization Structure

You can add workload domains to an organization in VCF Automation provider management for consumption by end users. You can create two types of organizations in VCF Automation - All Apps Organizations for workload domains with Supervisor clusters that run both containers and VMs, and a VM Apps Organization for workload domains with clusters that run only VMs.

Note:

You cannot use the same workload domain in an All Apps Organization and a VM Apps Organization. The restriction is also applied to the vCenter instance and the vSphere clusters of the workload domain.

1. [Deploy an NSX Edge cluster and set up network connectivity in the workload domain and create Virtual Private Cloud \(VPCs\) in vCenter.](#)
2. [Deploy a Supervisor with NSX VPC.](#)
3. [Add an identity provider for your organizations.](#)
4. [Create organizations for all applications for the workload domains with Supervisor instances.](#)
5. [Create an organization for VM applications for the workload domains that do not run Supervisor instances.](#)

Detailed Journeys

For visual guides to the sequences of getting started operations, see the following VCF journey maps.

- [Build Journey - Install a New VMware Cloud Foundation Deployment](#)
- [Build Journey - Converge Your Infrastructure to VMware Cloud Foundation from Existing vSphere and VMware Aria Operations](#)
- [Holistic Journey - Fully Adopt VMware Cloud Foundation from Existing vSphere and VMware Aria Operations](#)

Design Blueprints

You can apply a pre-defined architecture for your specific business and technology requirements. See [Design Blueprints for VMware Cloud Foundation](#).

List of VCF Capabilities

For a detailed list of capabilities and links to the documentation for them, see [Capabilities by Job Function](#).

What to Do Next

Share the details of the organizations you created in VCF Automation provider management with the administrators for those organizations. Organization administrators can start distributing the infrastructure and configuring access management to the groups in their organizations by using the organization management functionality in VCF Automation. See [Managing Organizations for All Apps in VCF Automation](#) and [Managing Organizations in VCF Automation for VM Apps](#).

Getting Started with vSphere Foundation

You can start building a vSphere Foundation private cloud by deploying a new platform or converging your existing virtual infrastructure to vSphere Foundation. After you deploy VCF Operations using a new or existing vCenter instance, you can continue with adding vCenter instances to the cloud.

In this article:

- [Deploy VCF Operations](#)
- [Add vCenter Instances to VCF Operations](#)
- [Complete the Configuration of vSphere Foundation](#)
- [Detailed Journeys](#)
- [What to Do Next](#)

Deploy VCF Operations

The first step in building a vSphere Foundation platform is deploying VCF Operations in one of the following ways:

- [Deploy VCF Operations with a new vCenter instance.](#)
- [Deploy VCF Operations on an existing vCenter instance.](#)

If you have an existing VMware Aria Operations and vCenter instance with managed vSphere clusters, upgrade VMware Aria Operations to VCF Operations 9.0, and vCenter and ESX to version 9.0.

Add vCenter Instances to VCF Operations

After you deploy or upgrade to VCF Operations, you can start adding vCenter instances with infrastructure running consumer workloads. See [vSphere as a data source in VCF Operations](#).

Complete the Configuration of vSphere Foundation

After you deploy VCF Operations or add a vCenter instance to it, complete the platform's configuration by performing the following tasks. Assigning a license must be done right after you complete deployment.

- [Assign a license.](#)
- [Configure backup of the management components.](#)
- [Configure certificate management in VCF Operations.](#)
- [Configure password management.](#)
- [Deploy and configure vSphere Supervisor.](#)

Detailed Journeys

For visual guides to the sequences of getting started operations, see the following vSphere Foundation journey maps.

- [Build Journey - Install a New vSphere Foundation Deployment](#)
- [Build Journey - Converge Existing vSphere to vSphere Foundation 9 with a New VCF Operations Instance](#)
- [Build - Upgrade Existing vSphere and VMware Aria Operations to vSphere Foundation 9](#)

What to Do Next

If you want to run Kubernetes workloads, VMs, or vSphere Pods on vSphere, see [Building Cloud Applications](#).